Deep Learning: Advanced Models and Methods (DL26)

Track 3: Graph-based Metric Learning for Scene Understanding

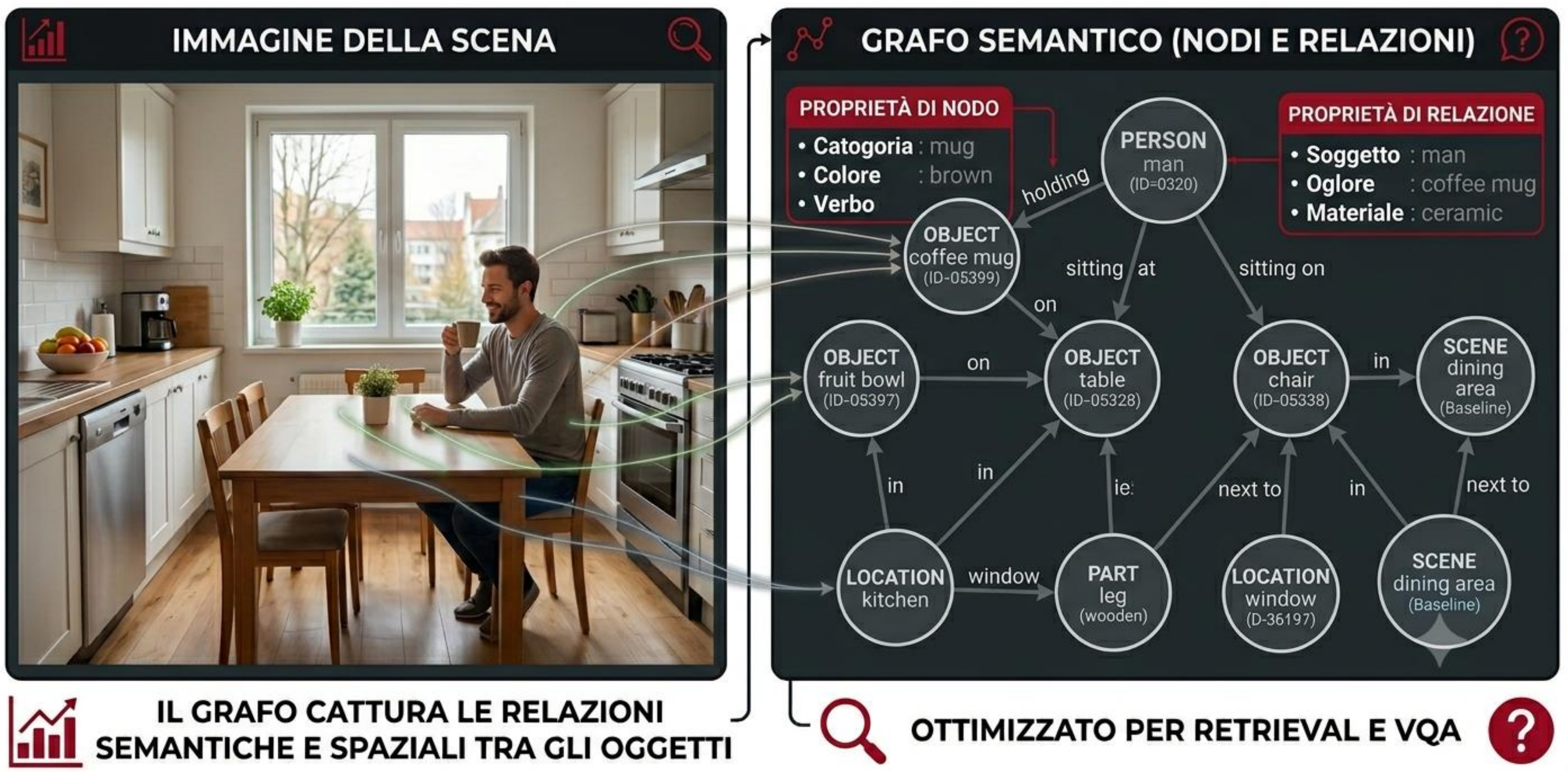
Gruppo: G18

Membri: [Salvatore Mario Carota]

Repository:

https://github.com/LenionK/dl26-projects

From Pixels to Graphs: Representing Scenes as Structured Data



GQA: Visual Reasoning in the Real World

2328363

2379201

2387152

2379505

2407089

2330388

2334735

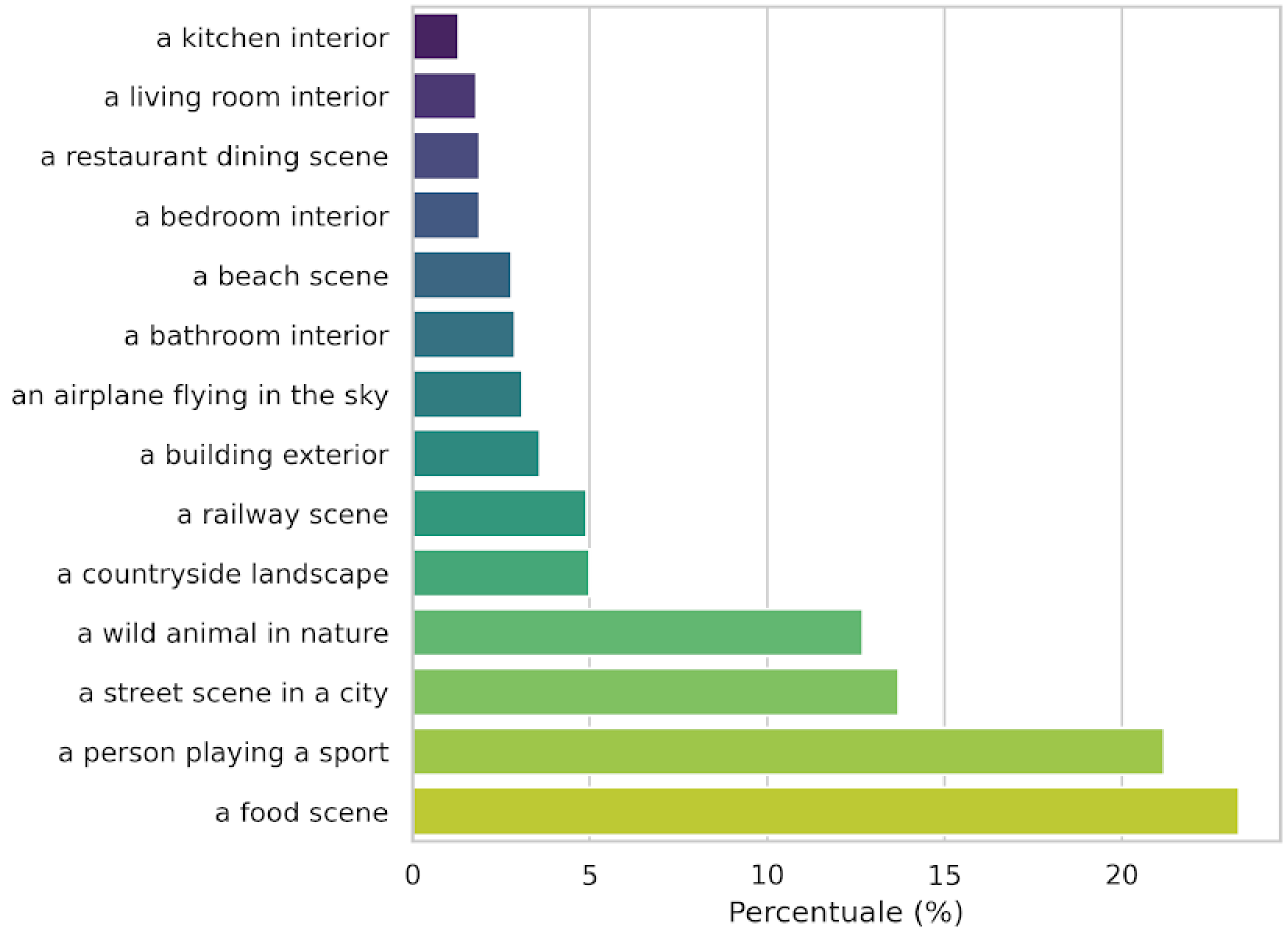
2327633

713232

2414195

Bridging the Gap: Automatic Labeling via CLIP

Classi di Partenza
Interni
Urbano
Natura
Cibo



Tre encoder, stesso obiettivo metrico

Baseline - Graph - Contrastive Training

Modello	Input	Architettura	Embedding
Baseline	Immagine	ResNet-18 → FC	128-d
GCN	Grafo	3× GCNConv + pool mean/max	128-d
GINE	Grafo + edge type	4× GINEConv + pool mean/max	128-d

Supervised Contrastive Loss

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\frac{1}{|P(i)|}\sum_{p\in P(i)}\log\left(\frac{\exp(z_i\cdot z_p/\tau)}{\sum_{j\neq i}\exp(z_j\cdot z_i/\tau)}\right)$$

Training e iperparametri

Parametro	Baseline	GCN/GINE
Optimizer	Adam (lr=1°-4)	Adam (lr=3e-4, weight_decay=1e-4)
Scheduler	-	CosineAnnealingLR
Grad clipping	-	Max_norm=1.0
Batch sampling	6 classi x 4 campioni	6 classi x 4 campioni
Epochs	10	20

Scene-To-Scene retrieval con metric learning

Retrieval Evaluation: Given a query scene graph, perform retrieval on the dataset to fetch structurally similar graphs, evaluating via standard classification metrics (Accuracy, Precision, Recall).

QUERY

CNN

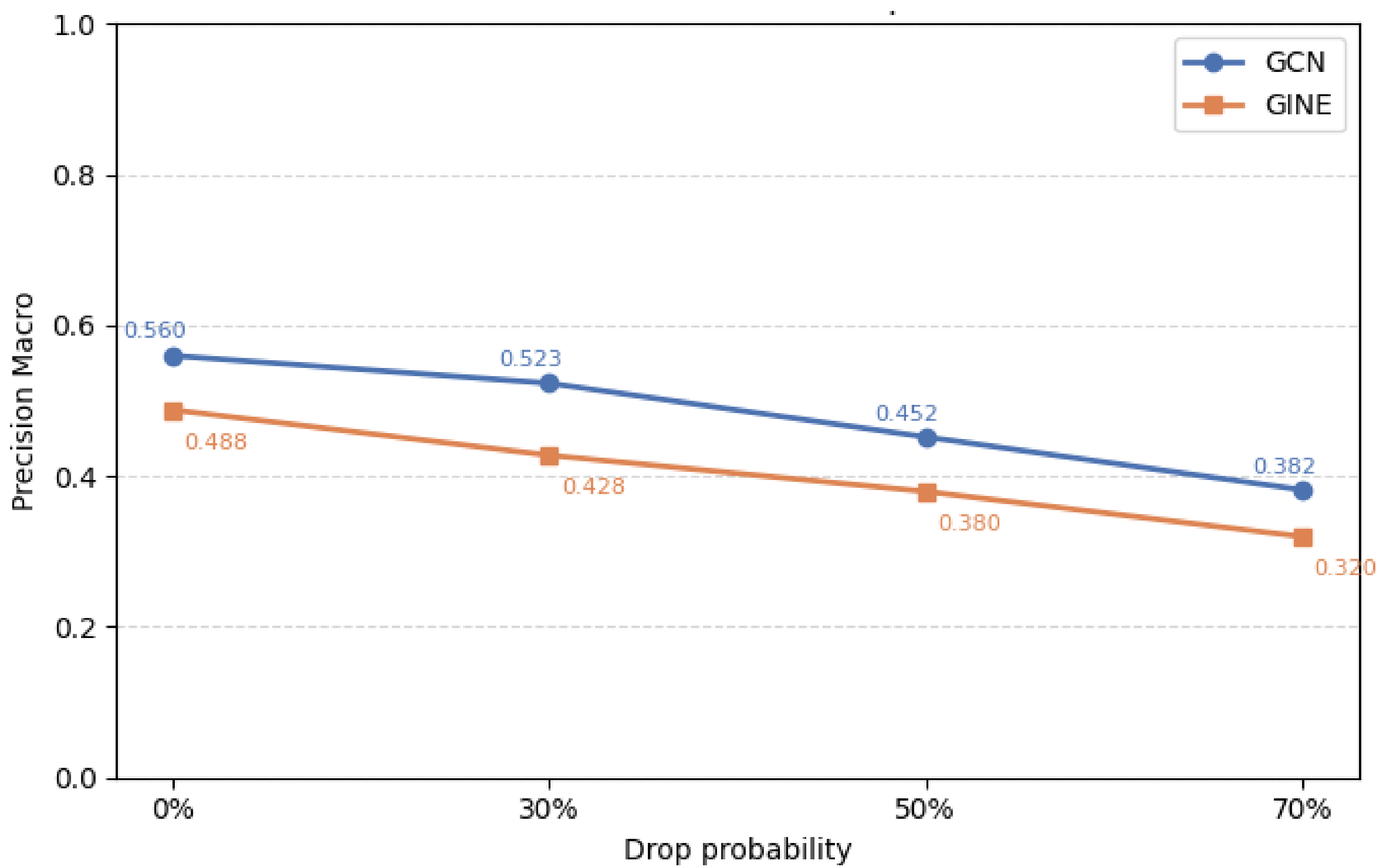
GCN

Metriche: Scene-To-Scene retrieval con metric learning

Metric	Baseline	GCN	GINE
acc@1	**0.6231**	0.5966	0.4991
precision_macro	**0.5855**	0.5673	0.4877
recall_macro	**0.5668**	0.5661	0.4892
recall@1	**0.6231**	0.5966	0.4991
recall@5	**0.8395**	0.8204	0.7883
recall@10	**0.8881**	0.8791	0.8572
precision@1	**0.6231**	0.5966	0.4991
precision@5	**0.6195**	0.6032	0.5002
precision@10	**0.6152**	0.6046	0.4978
mrr	**0.7163**	0.6949	621
map	**0.68**	0.6631	0.5778

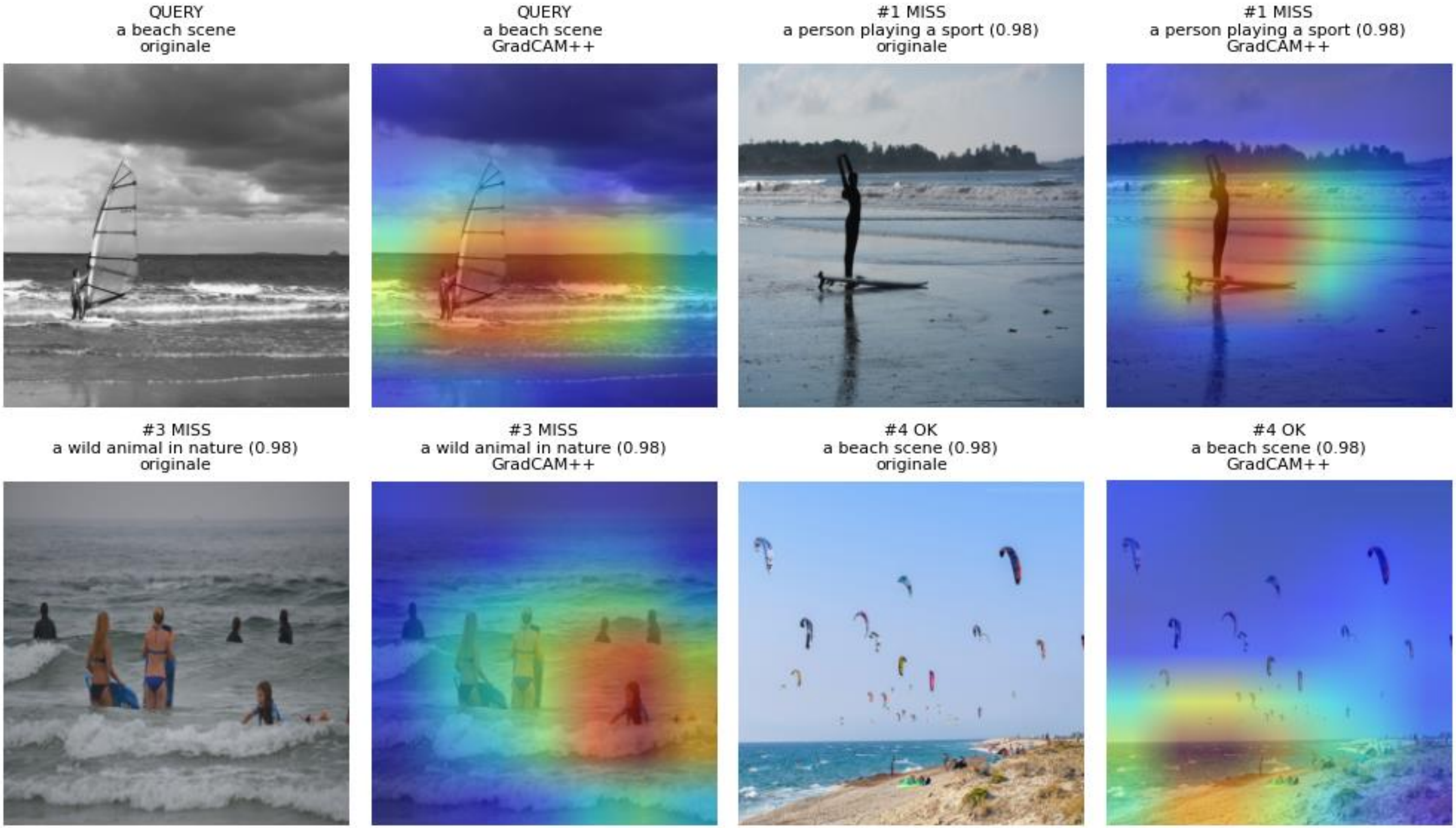
GNN degrada gradualmente anche con il 70% dei nodi rimossi

Extra Objective: robustezza alle perturbazioni strutturali



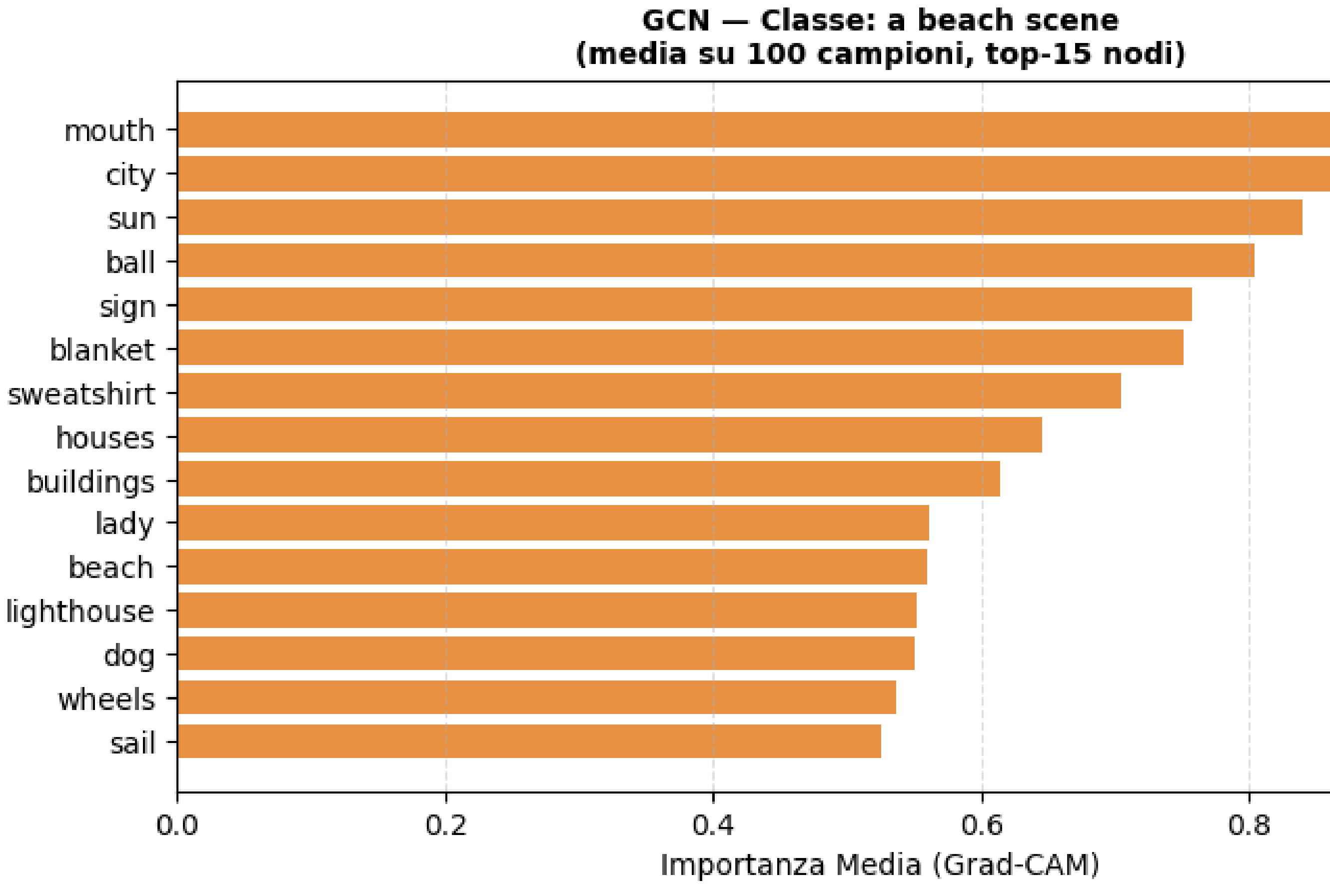
GRADCAM++

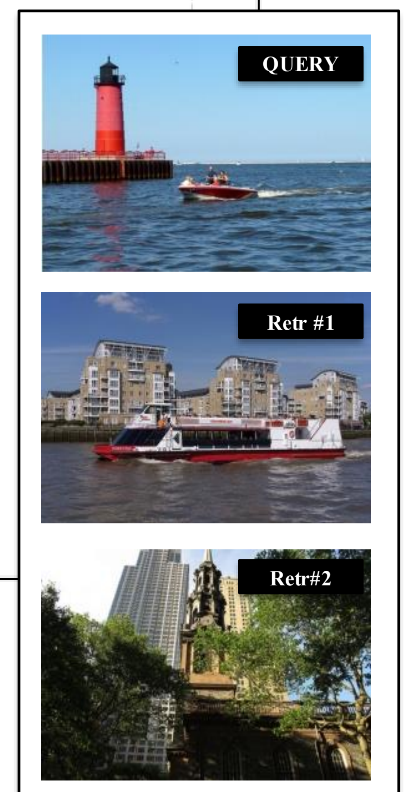
Extra Objective: Visualize and interpret which edges/relations act as critical focal points for the model's similarity metric.



GRAPHGRADCAM

Extra Objective





Focal Point Analysis — Ranking Disruption

Extra Objective: Visualize and interpret which edges/relations act as critical focal points for the model's similarity metric.

GCN

Relazioni	Disruption
leaves → tree	0.516667
person → shirt	0.400000
sky → clouds	0.366667
shirt → person	0.320000
clouds → sky	0.300000
coat → man	0.260000
tree → leaves	0.233333

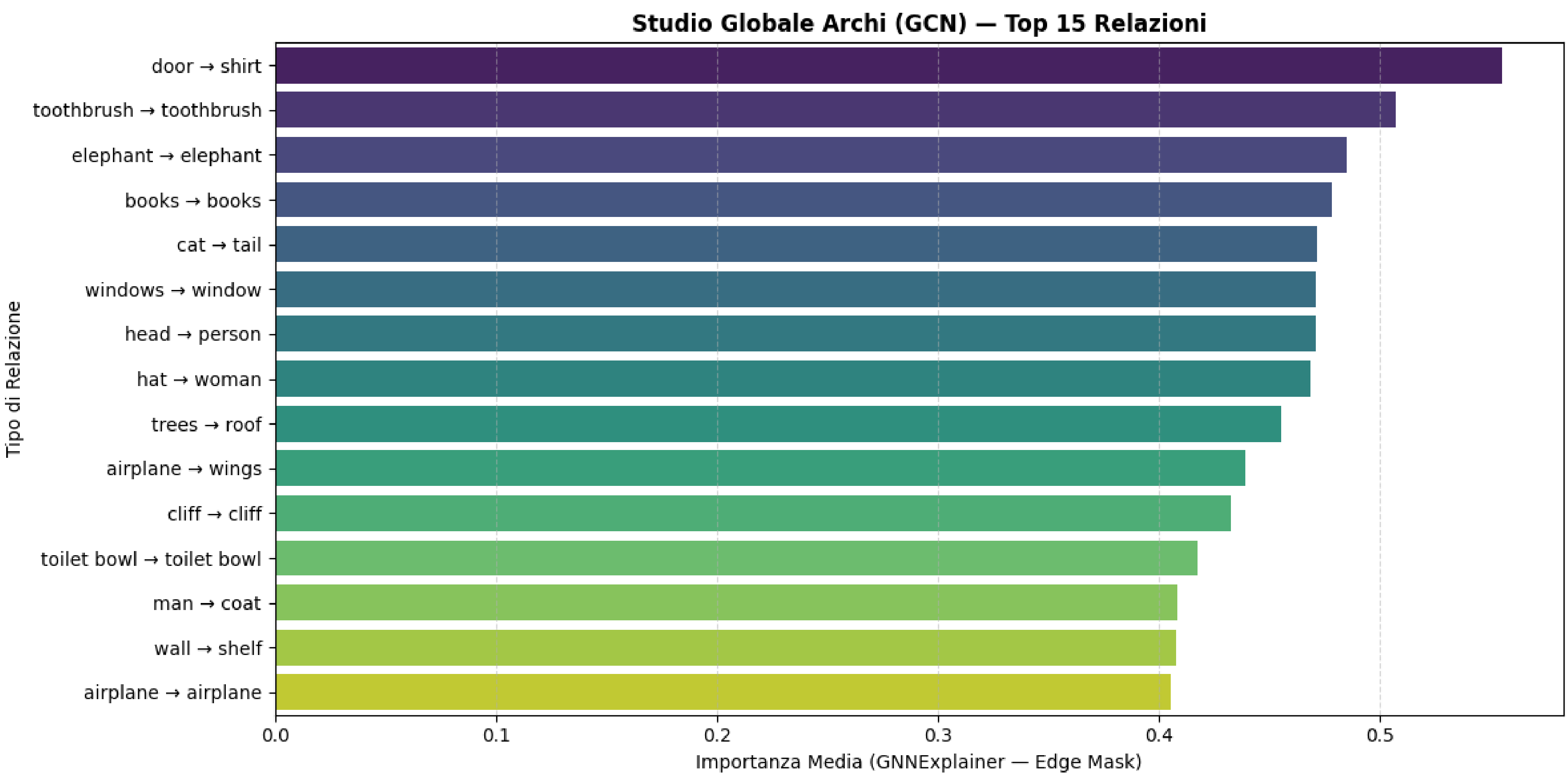
GINE

Relazioni	Disruption
to the left of	0.542857
to the right of	0.485714
on	0.245918
wearing	0.128571
in	0.121429
near	0.069388
of	0.063918

$$\text{Disruption} = 1 - \text{overlap}(\text{top-k base, top-k perturbed})$$
$$\text{overlap} = |\text{top-k}(\text{base}) \cap \text{top-k}(\text{perturbato})| \,/\, k$$

GCN: Studio Archi — GNNExplainer

Extra Objective: Visualize and interpret which edges/relations act as critical focal points for the model's similarity metric.



GINE: Studio Archi — GNNExplainer

Extra Objective: Visualize and interpret which edges/relations act as critical focal points for the model's similarity metric.